ESD217 Climate Change & Sustainability

MCQ QUESTIONS BANK

Part 1: Weather, Climate, and Energy Balance (Q1–Q30)

1. What is the primary defining difference between weather and climate?

A) Weather refers to long-term atmospheric patterns, while climate refers to daily changes.

B) Weather describes short-term changes in the atmosphere, whereas climate describes long-term patterns over typically 30 years or more in a specific area.

C) Weather only measures temperature, while climate measures precipitation and wind exclusively.

D) Weather and climate are interchangeable terms used to describe atmospheric states over identical time scales.

ANSWER: B

2. What term is used to refer to three-decade averages of weather observations?

A) Atmospheric Average

B) Meteorological Standard

C) Climate Baseline

D) Climate Normal

ANSWER: D

3. Why is the term "climate change" considered more accurate than "global warming"?

A) Because the Earth is actually cooling globally rather than warming.

B) Because climate change only refers to changes in precipitation, not temperature.

C) Because the climate does not warm uniformly around the globe; some areas like the Arctic warm much more, while precipitation changes cause varying droughts and storms.

D) Because global warming is a legally defined term restricted to industrial sectors.

ANSWER: C

4. Which of the following best exemplifies "weather" rather than "climate"?

A) The dry season in Kenya is from June to October.

B) Thailand has high humidity all year around.

C) It feels hotter than usual right now in Bali.

D) California historically experiences long, hot, dry summers.

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ANSWER: C

5. What does the term "anthropogenic" refer to?

A) The influence that humans can have on the natural world

B). Natural solar cycles affecting global temperatures.

C) Volcanic aerosol emissions during historical eruptions.

D) The natural greenhouse effect occurring without human intervention.

ANSWER: A

6. Approximately how much solar radiation (in watts per m^2) reaches the planet, and how much passes through the atmosphere?

A) 343 W/m^2 reaches the planet, and 240 W/m^2 passes through the atmosphere.

B) 440 W/m^2 reaches the planet, and 343 W/m^2 passes through the atmosphere.

C) 500 W/m^2 reaches the planet, and 100 W/m^2 passes through the atmosphere.

D) 1000 W/m^2 reaches the planet, and 500 W/m^2 passes through the atmosphere.

ANSWER: A

7. How does the energy absorbed from the Sun by Earth compare globally to the energy produced annually from fossil fuel combustion?

A) Earth absorbs three orders of magnitude more energy from the Sun than is produced globally from fossil fuel combustion over a half year.

B) Earth absorbs three orders of magnitude more energy from the Sun than is produced globally from fossil fuel combustion over an entire year.

C) Earth absorbs five orders of magnitude more energy from the Sun than is produced globally from fossil fuel combustion over a half year.

D) Earth absorbs five orders of magnitude more energy from the Sun than is produced globally from fossil fuel combustion over an entire year.

ANSWER: D

8. What is the primary region of the electromagnetic spectrum where most of the Sun's radiation and energy are concentrated?

A) Ultraviolet region

B) Infrared region

C) Visible region

D) X-ray region

ANSWER: C

9. Between ultraviolet (UV) and infrared (IR) radiation, which one has a shorter wavelength?

A) Infrared (IR) radiation

B) Ultraviolet (UV) radiation

C) Both have identical wavelengths

D) Neither possesses measurable wavelengths

ANSWER: B

10. When incident sunlight reaches Earth, what percentage is absorbed at the surface, absorbed by the atmosphere, and reflected back into space, respectively?

A) 70% absorbed at surface, 20% by atmosphere, 10% reflected

B) 50% absorbed at surface, 30% by atmosphere, 20% reflected

C) 60% absorbed at surface, 20% by atmosphere, 30% reflected

D) 50% absorbed at surface, 20% by atmosphere, 30% reflected

ANSWER: D

11. What is the term defined as the percentage of solar radiation reflected from a surface?

A) Emissivity

B) Albedo

C) Radiance index

D) Transmissivity

ANSWER: B

12. Which of the following surface types has the highest typical albedo value?

A) Fresh snow

B) Thick clouds

C) Asphalt

D) Forest

ANSWER: A

13. What is the global average albedo represented by the solar radiation reflected back into space?

A) 10%

B) 20%

C) 30%

D) 50%

ANSWER: C

14. According to blackbody radiation principles, how do temperature, energy intensity, and wavelength relate for radiating objects?

A) Hotter objects radiate less energy and at longer wavelengths than cooler objects.

B) Hotter objects radiate more energy, and at shorter wavelengths, than cooler objects.

C) Temperature has no effect on the wavelength of emitted radiation.

D) Cooler objects radiate more energy at shorter wavelengths than hotter objects.

ANSWER: B

15. What is the approximated blackbody temperature of the Sun, and where does its peak wavelength distribution fall?

A) 15°C with a peak in the infrared region around 10 micrometers.

B) 100°C with a peak in the microwave region.

C) $2,000^{\circ}$C with a peak in the ultraviolet region around 0.1 micrometers.

D) $5,500^{\circ}$C with a peak in the visible spectrum around 0.53 micrometers.

ANSWER: D

16. What is the average surface temperature of the Earth, and where is the peak of its emitted radiation located?

A) $5,500^{\circ}$C; visible spectrum

B) -19°C; ultraviolet region

C) 15°C; infrared region around 10 micrometers

D) 0°C; microwave region

ANSWER: C

17. Which of the following gases are the most abundant in the atmosphere yet are NOT classified as greenhouse gases?

A) Carbon dioxide and water vapor

B) Methane and ozone

C) Nitrogen and oxygen

D) Helium and argon

ANSWER: C

18. What is the technical physical mechanism by which greenhouse gas (GHG) molecules warm the atmosphere when hit by infrared radiation?

A) They absorb shortwave UV rays and convert them into visible light.

B) The GHG molecule starts vibrating when hit by infrared rays, which causes adjacent gas molecules of any kind to vibrate as well, converting vibration into heat energy per thermodynamics.

C) They chemically bond with oxygen molecules to release exothermic heat.

D) They reflect incoming solar radiation directly back to the Earth's surface.

ANSWER: B

19. By how much does the natural greenhouse effect raise the average world temperature on Earth?

A) Around $10^{\circ}C$

B) Around $24^{\circ}C$

C) Around $34^{\circ}C$

D) Around $50^{\circ}C$

ANSWER: C

20. Why is Venus extremely hot despite having a high albedo of 90%?

A) Because it is located much closer to the Sun than Mercury.

B) Because its atmosphere is 96.5% CO_2 with an atmospheric pressure 92 times greater than Earth's.

C) Because volcanic activity on Venus continually generates geothermal heat.

D) Because it has no atmosphere to reflect incoming solar radiation.

ANSWER: B

21. Why does Mars have a very weak greenhouse effect and a mean surface temperature very similar to its blackbody temperature ($-53^{\circ}C$ vs $-63^{\circ}C$)?

A) Because its atmosphere is composed entirely of oxygen and nitrogen.

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B) Because although its atmosphere is 95.3% CO_2, its atmospheric pressure is only $1/1{,}000$th that of Earth.

C) Because its albedo is 90%, reflecting almost all solar radiation.

D) Because it is located too close to the asteroid belt.

ANSWER: B

22. What is the blackbody temperature and actual surface temperature of Venus, respectively?

A) $-89°C$; $+477°C$

B) $-19°C$; $+15°C$

C) $-63°C$; $-53°C$

D) $+15°C$; $+100°C$

ANSWER: A

23. What are the blackbody and actual surface temperatures of Earth, respectively?

A) $-89°C$ and $+477°C$

B) $-19°C$ and $+15°C$

C) $-63°C$ and $-53°C$

D) $0°C$ and $+34°C$

ANSWER: B

24. What are the blackbody and actual surface temperatures of Mars, respectively?

A) $-19°C$ and $+15°C$

B) $-89°C$ and $+477°C$

C) $-63°C$ and $-53°C$

D) $-53°C$ and $-63°C$

ANSWER: C

25. What is insolation?

A) The amount of sunlight reaching the Earth.

B) The rate of heat loss from Earth into outer space.

C) The percentage of solar radiation reflected by clouds.

D) The vibration frequency of greenhouse gas molecules.

ANSWER: A

26. What percentage of solar radiation striking the Earth and its atmosphere is absorbed at the surface?

A) About 25%

B) About 50%

C) About 60%

D) About 70%

ANSWER: B

27. What is the temperature we feel on Earth a result of?

A) Direct friction from solar wind.

B) The balance of heat energy that is trapped on Earth by the planet's atmosphere.

C) Constant geothermal heating from Earth's core.

D) The combustion of fossil fuels globally.

ANSWER: B

28. What happens when there are more greenhouse gases in the atmosphere?

A) They reflect all incoming solar radiation immediately back into space.

B) They trap more heat, leading to a gradual increase in temperature and changes in climate.

C) They cool the stratosphere down to absolute zero.

D) They increase the atmospheric pressure on Venus.

ANSWER: B

29. What is the average temperature on Earth allowed by the thermal radiation absorbed and reflected back by greenhouse gases?

A) Around $14^{\circ}C$

B) Around $25^{\circ}C$

C) Around $-19^{\circ}C$

D) Around $100^{\circ}C$

ANSWER: A

30. Which of the following best describes blackbody radiators like the Sun and Earth?

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A) Objects whose emitted electromagnetic radiation intensity and wavelength depend on their temperature.

B) Objects that absorb 100% of light and emit zero radiation.

C) Objects that maintain a constant temperature regardless of external radiation.

D) Objects that only emit X-rays and gamma rays.

ANSWER: A

Part 2: Atmosphere Layers (Q31–Q60)

31. Which layer of the atmosphere is closest to the Earth's surface and contains the densest air where living organisms live?

A) Stratosphere

B) Mesosphere

C) Troposphere

D) Thermosphere

ANSWER: C

32. Up to what approximate altitude above Earth's surface does the troposphere extend?

A) 10 km

B) 15 km

C) 80 km

D) 90 km

ANSWER: A

33. How does air temperature change with elevation in the troposphere?

A) Temperature increases by 6.5°C for every 1,000 m of elevation.

B) Temperature decreases by 6.5°C for every 1,000 m of elevation.

C) Temperature remains completely constant throughout the troposphere.

D) Temperature increases exponentially up to the tropopause.

ANSWER: B

34. What is the function of the tropopause layer located at the top of the troposphere?

A) It contains the ozone layer that absorbs UV radiation.

B) It acts as a barrier or transitional zone that prevents the mixing of cold air from the troposphere with warm air in the stratosphere.

C) It reflects long-distance radio communications.

D) It allows satellites and spacecraft to orbit without air resistance.

ANSWER: B

35. What are the altitude boundaries of the stratosphere?

A) Surface to 10 km

B) 10 km to 50 km

C) 20 km to 50 km

D) 10 km to 60 km

ANSWER: B

36. Why does temperature start to rise in the upper portion of the stratosphere?

A) Due to the presence of the ozone layer, which absorbs UV from solar radiation.

B) Due to the absorption of high-frequency X-rays by nitrogen and oxygen.

C) Because it is closest to the Sun.

D) Because of intense volcanic ash suspension.

ANSWER: A

37. Why is the stratosphere considered the preferred layer for aircraft flying?

A) Because the air is densest there, providing maximum lift.

B) Because air speed in it is low.

C) Because air currents movement in it is vertical.

D) Because air currents movement in it is horizontal.

ANSWER: D

38. What are the altitude boundaries and temperature characteristics of the mesosphere?

A) 50 to 80 km; temperatures rise to $2,000°C$

B) 50 to 80 km; it is the coldest layer of the atmosphere, reaching $-93°C$ at its top due to low gas molecules capable of absorbing solar radiation.

C) 80 to 700 km; temperatures drop to absolute zero.

D) 700 to 10,000 km; molecules achieve tremendous speeds.

ANSWER: B

39. What is the primary importance of the mesosphere for life on Earth?

A) Most meteors falling from space burn up and disintegrate after entering the Earth's atmosphere as they pass through this layer.

B) It reflects radio waves for global communication.

C) It absorbs all infrared radiation emitted by Earth's surface.

D) It houses the ozone layer protecting against UV rays.

ANSWER: A

40. What are the altitude boundaries and temperature characteristics of the thermosphere?

A) 80 km to 700 km; temperatures reach higher than $2,000°C$ because oxygen and nitrogen absorb high-frequency solar radiation like X-rays and gamma rays.

B) 50 km to 80 km; temperatures drop to $-93°C$.

C) 700 km to 10,000 km; temperatures remain below freezing.

D) 80 km to 700 km; temperatures reach higher than $1500°C$ because oxygen and nitrogen absorb high-frequency solar radiation like X-rays and gamma rays.

ANSWER: A

41. What is the name and altitude range of the lower part of the thermosphere that is used in long-distance radio communications?

A) Tropopause; 10 to 11 km

B) Ionosphere; 80 km to 550 km

C) Ionosphere; 70 km to 400 km

D) Stratopause; 40 to 50 km

ANSWER: B

42. What are the altitude boundaries and distinguishing characteristics of the exosphere?

A) 50 to 80 km; coldest layer where meteors burn.

B) 80 to 700 km; home of the ionosphere reflecting radio waves.

C) 550 km to 10,000 km; highest layer where atoms and molecules move over great distances without colliding, achieving speeds that allow them to escape Earth's gravity.

D) 700 km to 10,000 km; highest layer where atoms and molecules move over great distances without colliding, achieving speeds that allow them to escape Earth's gravity.

ANSWER: D

43. Why is the exosphere considered the ideal region for placing satellites and spacecraft?

A) Because the extreme heat destroys space debris.

B) Because they face almost no resistance due to extremely high air density, allowing them to remain in orbits for long periods.

C) Because they face almost no resistance due to extremely low air density, allowing them to remain in orbits for long periods.

D) Because oxygen and nitrogen concentrations provide necessary combustion.

ANSWER: C

44. What are the essential gases in the troposphere that provide fundamental components necessary for life?

A) Oxygen, carbon dioxide and other gases

B) Oxygen, carbon dioxide, Ozone, and other gases

C) Ozone, sulfur dioxide, and nitrogen oxides

D) Oxygen, carbon dioxide, water vapor, and other gases

ANSWER: D

45. What is the average surface temperature at Earth's surface in the troposphere?

A) $0^{\circ}C$

B) $15^{\circ}C$

C) $25^{\circ}C$

D) $-15^{\circ}C$

ANSWER: B

46. In which atmospheric layer do weather and climate phenomena such as cloud formation, rainfall, and wind movement primarily occur?

A) Stratosphere

B) Mesosphere

C) Troposphere

D) Thermosphere

ANSWER: C

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47. What specific solar radiation types are absorbed by oxygen and nitrogen gases in the thermosphere?

A) Visible light and infrared radiation

B) High-frequency solar radiation such as X-rays and gamma rays

C) Low-frequency solar radiation such as X-rays and gamma rays

D) Radio waves and microwaves

ANSWER: C

48. Which atmospheric layer acts as a barrier preventing cold air mixing from the troposphere with warm air in the stratosphere?

A) Ionosphere

B) Tropopause

C) Stratopause

D) Mesopause

ANSWER: B

49. What is the approximate altitude of the tropopause layer?

A) 10 kilometers

B) 11 kilometers

C) 30 kilometers

D) 80 kilometers

ANSWER: B

50. Which layer extends from an altitude of 10 km to about 50 km?

A) Troposphere

B) Thermosphere

C) Mesosphere

D) Stratosphere

ANSWER: D

51. What is the temperature at the top of the mesosphere?

A) $-53^{\circ}C$

B) $-93^{\circ}C$

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C) $+15°C$

D) $+2,000°C$

ANSWER: B

52. What is the lower altitude limit of the ionosphere within the thermosphere?

A) 10 km

B) 50 km

C) 70 km

D) 80 km

ANSWER: D

53. What is the upper altitude limit of the ionosphere?

A) 550 km

B) 700 km

C) 1,000 km

D) 10,000 km

ANSWER: A

54. Which layer is known as the highest layer in the atmosphere?

A) Thermosphere

B) Exosphere

C) Mesosphere

D) Stratosphere

ANSWER: B

55. What is the maximum altitude of the exosphere mentioned in the atmosphere layers overview?

A) 700 km

B) 1,000 km

C) 5,500 km

D) 10,000 km

ANSWER: D

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56. At what altitude in the stratosphere does the temperature start to rise due to ozone absorption?

A) Above 20 km

B) At exactly 10 km

C) At 50 km

D) At 80 km

ANSWER: A

57. Based on the atmospheric layers diagram, which layer lies immediately above the mesosphere?

A) Stratosphere

B) Troposphere

C) Thermosphere

D) Exosphere

ANSWER: C

58. Which atmospheric layer is characterized by air density being highest at the bottom and decreasing rapidly with height?

A) Troposphere

B) Exosphere

C) Thermosphere

D) Mesosphere

ANSWER: A

59. What is the primary reason why temperatures in the thermosphere exceed $2,000^{\circ}$C?

A) Low air pressure traps geothermal heat.

B) Oxygen and nitrogen gases absorb low-frequency solar radiation like X-rays and gamma rays.

C) Proximity to the Earth's molten core.

D) Oxygen and nitrogen gases absorb high-frequency solar radiation like X-rays and gamma rays.

ANSWER: D

60. What physical characteristic allows satellites in the exosphere to remain in orbit for long periods?

A) Extremely high air density resulting in almost no resistance.

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B) Extremely low air density resulting in almost no resistance.

C) High concentration of ozone molecules providing lift.

D) Constant solar wind propulsion.

ANSWER: A

Part 3: Coral Reef Bleaching & Climate History (Q61–Q105)

61. What microscopic algae live in the tissues of corals in a symbiotic relationship, providing their primary food source and color?

A) Phytoplankton

B) Zooxanthellae

C) Diatoms

D) Cyanobacteria

ANSWER: B

62. What is the leading cause of coral bleaching according to NOAA?

A) Ocean acidification caused by sulfur dioxide

B) Increased ocean temperature caused by climate change

C) Overfishing of herbivorous fish species

D) Increased salinity from melting polar ice

ANSWER: B

63. What happens to corals during the bleaching process when they become stressed due to increased ocean temperature or pollution?

A) They absorb excess zooxanthellae and turn dark brown.

B) The algae leave the coral's tissue, causing the coral to lose its major source of food, turn white or very pale, and become more susceptible to disease.

C) Their calcium carbonate skeletons dissolve completely into liquid.

D) They rapidly migrate to deeper, colder waters.

ANSWER: B

64. Besides increased ocean temperature, what other factors can cause coral bleaching in shallow-water corals?

A) Runoff and pollution from storm-generated precipitation, overexposure to sunlight, and exposure to air during extreme low tides.

B) Complete absence of sunlight and high sub-zero water temperatures.

C) Excessive volcanic ash deposition and high atmospheric pressure.

D) High levels of dissolved oxygen and nitrogen gas.

ANSWER: A

65. How does storm-generated precipitation contribute to bleaching in near-shore corals?

A) By increasing the salinity of ocean water to toxic levels.

B) By rapidly diluting ocean water and carrying pollutants via runoff.

C) By lowering water temperatures below freezing.

D) By blocking all solar irradiance permanently.

ANSWER: B

66. Which of the following are cited as examples of anthropogenic climate change impacts on wildlife in Egypt?

A) Ostrich and seals

B) Ostrich and Ghazal

C) Wryneck and penguins

D) Coral reefs and Ghazal

ANSWER: B

67. Which of the following are cited as examples of worldwide anthropogenic climate change impacts?

A) Ostrich and Gazal

B) Polar bear, seal, and Gazal

C) Desert camels and scorpions

D) Polar bear, seal, and Eurasian Wryneck

ANSWER: D

68. What are the three primary factors that determine the Earth's climate over long-time scales?

A) Tidal forces, lunar eclipses, and continental drift.

B) Solar input, the concentration of greenhouse gases, and the average global albedo.

C) Ocean currents, atmospheric pressure, and cloud cover.

D) Volcanic ash density, meteor impacts, and seismic activity.

ANSWER: B

69. What is the field of study known as that looks back in time at ancient climate using materials that retained unique signatures?

A) Meteorology

B) Paleoclimatology

C) Stratigraphy

D) Seismology

ANSWER: B

70. What are paleoclimate proxies?

A) Direct thermometer measurements recorded over the last 200 years.

B) Materials that have retained a unique signature of ancient climate conditions, such as ice cores, tree rings, and pollen spores.

C) Artificial computer models simulating future climate scenarios.

D) Historical documents written exclusively during the Industrial Revolution.

ANSWER: B

71. Prior to 2.6 million years ago during pre-Quaternary times, how did global temperatures and CO_2 levels generally compare to today?

A) Temperatures and CO_2 levels were both much lower than they are now.

B) Global temperatures were generally higher than they are now, typically correlated with atmospheric CO_2 levels that were much greater than today.

C) Temperatures fluctuated wildly every 11 years with sunspot cycles.

D) CO_2 levels were zero, and temperatures matched modern icehouse conditions.

ANSWER: B

72. During the Early Eocene (52–48 million years ago), what were the atmospheric CO_2 concentrations and temperature conditions?

A) CO_2 concentrations were below 100 ppm and temperatures were freezing.

B) CO_2 concentrations were above 1,000 parts per million (ppm)—more than three times as high as modern preindustrial periods—and temperatures were higher.

C) CO_2 levels matched modern preindustrial levels exactly.

D) CO_2 levels reached 10,000 ppm, causing mass extinction.

ANSWER: B

73. What was the Paleocene-Eocene Thermal Maximum (PETM)?

A) A 100,000-year ice age caused by massive volcanic ash clouds.

B) A brief period at the boundary between the Paleocene and Eocene lasting only a few tens of thousands of years featuring a massive release of carbon into the atmosphere that precipitated a large increase in global temperatures.

C) A period when the Sun's output dropped by 50%.

D) The transition from the Holocene to the Pleistocene epoch.

ANSWER: B

74. What are stomata in plants, and how are they used in paleoclimatology?

A) Roots that absorb calcium carbonate; used to measure ocean salinity.

B) Pores in leaves that regulate the flow of CO_2 and H_2O; their density in ancient leaf fossils is used to recreate ancient atmospheric CO_2 concentrations.

C) Growth rings in tree trunks; used to determine annual precipitation.

D) Bacterial shells in seafloor sediments; used to measure oxygen isotopes.

ANSWER: B

75. How does stomata density correlate with atmospheric CO_2 levels in plants?

A) High atmospheric CO_2 results in high stomata density.

B) High atmospheric CO_2 results in low stomata density, while low atmospheric CO_2 indicates high stomata density.

C) Stomata density has no relationship with atmospheric CO_2.

D) Low atmospheric CO_2 causes plants to have zero stomata.

ANSWER: B

76. What is the Phanerozoic eon ("visible life"), and how long ago does it span?

A) The last 10,000 years of human civilization.

B) The last 541 million years, during which Earth's temperature shifted gradually between icehouse and hothouse conditions.

C) The 2 billion years of Precambrian time.

D) The 50-year period following the Industrial Revolution.

ANSWER: B

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77. When did permanent ice sheets first appear at the South Pole, ushering in Cenozoic icehouse conditions?

A) About 2 million years ago

B) About 34 million years ago

C) About 145.5 million years ago

D) About 10,000 years ago

ANSWER: B

78. What replaced the lush greenhouse forests (like palm trees in Wyoming and banana trees in Oregon) in North America as the climate cooled?

A) Tundra and permafrost

B) Open grasslands, which are more suitable for a colder, drier icehouse climate

C) Dense mangrove swamps

D) Hyper-arid sand deserts

ANSWER: B

79. What marked the beginning of the Quaternary epoch about 2 million years ago?

A) The extinction of the dinosaurs.

B) Earth's climate being sufficiently frigid to sustain massive ice sheets at both poles, with ice moving far south in the Northern Hemisphere.

C) The invention of agriculture by human civilization.

D) The Paleocene-Eocene Thermal Maximum.

ANSWER: B

80. How frequently has the cycle of glacial ice advance and retreat been happening for the last 800,000 years?

A) Roughly every 11 years

B) Roughly every 1,000 years

C) Roughly every 100,000 years

D) Roughly every 50 million years

ANSWER: C

81. When did the last significant ice sheet advance peak?

A) Approximately 2 million years ago

B) Approximately 18,000 years ago

C) Approximately 7,000 years ago

D) Approximately 1783 AD

ANSWER: B

82. Why are seafloor sediments useful recorders of worldwide climate change?

A) Because they contain petroleum reserves trapped during the Cenozoic.

B) Because the numbers and types of marine organisms living near the sea surface change with climate, and their dead shells sink to the ocean floor to form a sedimentary record.

C) Because ocean floor temperatures never change over millions of years.

D) Because deep-sea sediments absorb cosmic rays directly from space.

ANSWER: B

83. What does oxygen isotope analysis in deep-sea sediment shells and glacial ice rely upon?

A) Measuring the ratio of carbon-14 to carbon-12.

B) Measuring the exact ratio of oxygen isotopes, specifically $^{18}O/^{16}O$.

C) Measuring the concentration of dissolved nitrogen gas.

D) Measuring the radioactivity of uranium-lead decay.

ANSWER: B

84. What does a greater ratio of $^{18}O/^{16}O$ in marine shells indicate?

A) A period when ice sheets were expanding.

B) A hothouse period with zero ice sheets.

C) A massive volcanic eruption of sulfur dioxide.

D) High sunspot activity during the Maunder minimum.

ANSWER: A

85. How does temperature affect ^{18}O evaporation and precipitation?

A) When temperatures are high, more ^{18}O evaporates from oceans; when temperatures are low, less evaporates, making the heavy isotope less prevalent during colder times and more prevalent during warm eras.

B) Temperature has no effect on oxygen isotope evaporation rates.

C) Colder temperatures cause more ^{18}O to evaporate than warm temperatures.

D) ^{18}O only exists in solid rock, never in water vapor.

ANSWER: A

86. How long can vertical ice cores extracted from Antarctic and Greenland ice sheets be, and how many years of climate history might they include?

A) Up to 100 meters long; 1,000 years of history

B) Up to 2,000 meters (6,500 feet) long; over 200,000 years of climate history

C) Up to 10 kilometers long; 50 million years of history

D) Up to 50 meters long; 500 years of history

ANSWER: B

87. Besides oxygen isotope data, what additional climate records are preserved in glacial ice cores?

A) Tree ring widths and marine fish populations

B) Air bubbles recording atmospheric gas composition (such as CO_2 and methane), volcanic ash, wind-blown dust, pollen, and contemporary pollution.

C) Human historical documents and agricultural crop yields.

D) Stomata density from land plants exclusively.

ANSWER: B

88. What do tree growth rings reveal about historical climates?

A) The thickness and density of tree rings indicate environmental conditions (precipitation and temperature) during each year of growth.

B) The magnetic field orientation of the Earth at the time of growth.

C) The exact date of volcanic eruptions through ash layers embedded in the bark.

D) The concentration of ozone in the stratosphere.

ANSWER: A

89. Why are fossil pollen and spores valuable for reconstructing past climates?

A) Because plants produce pollen only during icehouse periods.

B) Because climate is a major factor influencing vegetation distribution, and pollen/spores have very resistant walls that make them abundant and well-preserved in sediments.

C) Because pollen grains act as direct thermometers in ocean water.

D) Because pollen records atmospheric pressure changes directly.

ANSWER: B

90. How do corals function as a "paleothermometer"?

A) They measure air temperature in the thermosphere using satellite telemetry.

B) They extract calcium carbonate ($CaCO_3$) from seawater to build skeletons containing oxygen isotopes reflecting water temperature, and exhibit seasonal growth bands due to density variations.

C) They migrate to different latitudes when water temperatures change by 1°C.

D) They burn up when exposed to ultraviolet radiation.

ANSWER: B

91. What can oxygen isotope analysis of coral-growth rings serve as a proxy measurement for in areas with large annual rainfall variations?

A) Wind speed

B) Precipitation

C) Atmospheric pressure

D) Solar irradiance

ANSWER: B

92. Why are historical documents sometimes challenging to use for precise climate analysis?

A) Because ancient manuscripts were written in languages that cannot be translated.

B) Because most manuscripts were written for purposes other than climate description, neglecting periods of stable atmospheric conditions and mentioning only extremes like droughts, severe storms, and blizzards.

C) Because historical documents are always falsified or exaggerated.

D) Because they only contain records from the 20th century.

ANSWER: B

93. What do harvest dates for the beginning of the grape harvest in the fall provide in European historical records?

A) A measure of volcanic sulfur dioxide emissions.

B) An integrated measure of temperature and precipitation during the growing season, providing a useful record of year-to-year climate variations.

C) The exact date of sunspot maximums.

D) Evidence of prehistoric ice sheet advances.

ANSWER: B

94. Over the past 2 billion years, how has the Earth's climate generally alternated?

A) Between a frigid "Ice House" and a steaming "Hothouse"

B) Between permanent tropical rainforests and hyper-arid deserts

C) Between constant glaciation and total ocean evaporation

D) Between stable Holocene conditions and Venus-like runaway greenhouse states

ANSWER: A

95. What characterizes the Holocene epoch (the last 10,000 years until now)?

A) Uniquely stable and warm conditions with sea level stabilizing about 7,000 years ago, allowing human civilization to develop.

B) Continuous ice sheet expansion covering all of North America.

C) Massive volcanic eruptions occurring every 11 years.

D) Extreme temperature fluctuations exceeding $50^{\circ}C$ annually.

ANSWER: A

96. What was the Little Ice Age?

A) The glacial maximum occurring 18,000 years ago.

B) A temperature decline and especially cold period in Europe between about 1600 and 1850.

C) The Paleocene-Eocene Thermal Maximum.

D) The Eocene greenhouse period 50 million years ago.

ANSWER: B

97. What is the term for factors that may influence Earth's climate over long periods of time (e.g., solar variability, volcanic activities, carbon cycle changes)?

A) Climate feedback loops

B) Climate forcing

C) Atmospheric damping

D) Radiative stabilization

ANSWER: B

98. What is the primary composition of hard coral skeletons extracted from seawater?

A) Calcium sulfate

B) Calcium carbonate

C) Silicon dioxide

D) Sodium chloride

ANSWER: B

99. Where was the 350-year sea-surface temperature record obtained through oxygen isotope analysis of coral?

A) Red Sea

B) Great Barrier Reef

C) Galapagos Islands

D) Mediterranean Sea

ANSWER: C

100. What causes the density variations in coral skeletons that form seasonal growth bands?

A) Variations in growth rates related to temperature and other environmental factors between winter and summer.

B) Lunar gravitational tides changing weekly.

C) Periodic volcanic ash deposition on reefs.

D) Variations in ocean salinity caused by meteors.

ANSWER: A

101. Why do historical documents often neglect periods of stable atmospheric conditions?

A) Because writers only cared about financial accounting.

B) Because writers understandably neglected normal conditions and mentioned only droughts, severe storms, memorable blizzards, and other extremes.

C) Because ink was too expensive to record normal weather.

D) Because stable weather never occurred historically.

ANSWER: B

102. What does the term "Phanerozoic" mean?

A) "Hidden life"

B) "Visible life"

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C) "Ancient ice"

D) "Warm greenhouse"

ANSWER: B

103. How many times in the last 541 million years has Earth's temperature shifted gradually between icehouse and hothouse conditions?

A) Countless times every 11 years

B) Only a few times

C) Exactly 50 times

D) Never; climate has always been identical

ANSWER: B

104. What geological era preceded the Cenozoic era and featured greenhouse climates experienced by dinosaurs?

A) Paleozoic era

B) Mesozoic era

C) Precambrian eon

D) Quaternary period

ANSWER: B

105. How far south in the Northern Hemisphere did ice move during the Quaternary ice advances before retreating to Greenland?

A) As far south as the modern-day Ohio River

B) As far south as the Gulf of Mexico

C) Only as far as the Arctic Circle

D) As far south as the Equator

ANSWER: A

Part 4: Climate Forcing, Solar Variability, and Volcanism (Q106–Q150)

106. How much solar variation has been experienced in the past 140 years according to satellite and solar measurements?

A) Very massive variations exceeding 50% total energy output.

B) Very little solar variation.

C) A steady linear increase of 10% per decade.

D) Complete cessation of solar radiation during winter months.

ANSWER: B

107. What are sunspots, and what phenomenon are they associated with?

A) Cold ocean currents; associated with hurricane formation.

B) Huge magnetic storms extending from the Sun's surface deep into the interior, associated with the ejection of particles that produce the aurora borealis (Northern Lights).

C) Volcanic craters on the lunar surface; associated with Earth tides.

D) Ozone depletion holes in the stratosphere; associated with UV radiation.

ANSWER: B

108. What is the approximate cycle length of sunspot activity (number of sunspots reaching a maximum)?

A) Every 5 years

B) Every 11 years

C) Every 100,000 years

D) Every 50 million years

ANSWER: B

109. Based on space measurements since 1978, what is the magnitude of solar variation during an 11-year sunspot cycle?

A) About 0.1%

B) About 5%

C) About 25%

D) About 50%

ANSWER: A

110. What was the Maunder minimum (1645–1715)?

A) A period of maximum sunspot activity corresponding with global hothouse conditions.

B) A period during which sunspots were largely absent, closely corresponding with the Little Ice Age cold period in Europe.

C) A period of massive volcanic eruptions across Indonesia.

D) The time when sea levels rose by 200 meters.

ANSWER: B

111. How do large explosive volcanic eruptions affect the Earth's surface temperature?

A) They warm the troposphere by releasing large amounts of carbon dioxide and water vapor.

B) They inject huge quantities of gases (like SO_2) and fine-grained debris/aerosols high into the atmosphere, which filter out incoming solar radiation and cool the earth's surface.

C) They increase global albedo by melting polar ice sheets.

D) They destroy the ozone layer in the troposphere.

ANSWER: B

112. What event followed the 1815 eruption of Mount Tambora in Indonesia, the largest eruption of modern times?

A) The Paleocene-Eocene Thermal Maximum

B) The "year without a summer" in 1816, with heavy snow in June and frost in July/August across northeastern US, Canada, and Western Europe.

C) A 10-year drought across the Middle East.

D) The complete melting of Arctic sea ice.

ANSWER: B

113. Which three more recent volcanic events have provided considerable data and insight regarding volcanic impacts on global temperatures?

A) Mount Etna (2002), Krakatoa (1883), and Vesuvius (79 AD)

B) Mount Tambora (1815), Laki (1783), and Santorini (1600 BC)

C) Mauna Loa (2010), Eyjafjallajökull (2010), and Kilauea (2018)

D) Mount St. Helens (1980), El Chichón (1982), and Mount Pinatubo (1991)

ANSWER: D

114. What is the Cretaceous period often called, and when did it occur?

A) The "age of mammals"; 20 million years ago to present.

B) The "age of dinosaurs", beginning about 145.5 million years ago and ending about 65.5 million years ago.

C) The Quaternary epoch; 2 million years ago to 10,000 years ago.

D) The Precambrian eon; 2 billion years ago.

ANSWER: B

115. What characterized the climate of the Cretaceous period?

A) Frigid icehouse conditions with massive ice sheets covering the equator.

B) Among the warmest in Earth's history, with dinosaurs ranging north of the Arctic Circle, tropical forests in Greenland and Antarctica, and sea level up to 200 meters higher than today.

C) Frequent volcanic winters with heavy snow in June.

D) Stable Holocene-like temperatures supporting human civilization.

ANSWER: B

116. What is considered the probable primary source of the additional CO_2 that caused the unusually warm Cretaceous hothouse climate?

A) Anthropogenic fossil fuel combustion by prehistoric civilizations.

B) An unusually high rate of volcanic activity, including massive outpourings of lava from oceanic lava plateaus produced by mantle plumes/hot spots.

C) Sudden changes in Earth's orbital eccentricity.

D) Complete evaporation of the world's oceans.

ANSWER: B

117. How is "climate" formally defined?

A) The daily mix of weather events in the atmosphere changing over minutes or hours.

B) The "average weather", or the statistical description in terms of mean and variability of relevant quantities over a period ranging from months to thousands or millions of years.

C) The exact atmospheric pressure at sea level during the summer solstice.

D) The total amount of solar radiation absorbed by land surfaces annually.

ANSWER: B

118. How is "weather" formally defined?

A) A 30-year average of precipitation and temperature observations.

B) The mix of events that happen each day in our atmosphere, varying over minutes, hours, days, and weeks.

C) The long-term geological evolution of atmospheric composition.

D) The baseline temperature control system of the planet.

ANSWER: B

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119. Which analogy is used to distinguish weather from climate regarding football?

A) "Weather is the referee, climate is the stadium."

B) "Weather is one football game, climate is the history of the National Football League".

C) "Weather is the football, climate is the playbook."

D) "Weather is the halftime show, climate is the championship trophy."

ANSWER: B

120. What does climate tell you versus what weather tells you, according to the clothing analogy?

A) Climate tells you what clothes to buy; weather tells you what clothes to wear.

B) Climate tells you what clothes to wash; weather tells you where to store them.

C) Climate tells you what shoes to wear; weather tells you when to sleep.

D) Climate and weather both dictate identical clothing choices.

ANSWER: A

121. What is the driving force behind the global temperature increase observed since before the Industrial Revolution?

A) Increasing solar output from 11-year sunspot cycles.

B) Increasing concentrations of greenhouse gases in the atmosphere due to human activities.

C) Decreasing global albedo caused by volcanic ash.

D) Expansion of open grasslands in North America.

ANSWER: B

122. What percentage of incident sunlight is absorbed by the Earth's atmosphere?

A) 10%

B) 20%

C) 30%

D) 50%

ANSWER: B

123. What percentage of incident sunlight is absorbed at the Earth's surface?

A) 20%

B) 30%

C) 50%

D) 70%

ANSWER: C

124. What percentage of incident sunlight is reflected back into space by the atmosphere, clouds, and reflective surface areas?

A) 20%

B) 30%

C) 50%

D) 80%

ANSWER: B

125. Which surface albedo range corresponds to fresh snow?

A) 5–10%

B) 10–20%

C) 35–50%

D) 80–95%

ANSWER: D

126. Which surface albedo range corresponds to asphalt?

A) 5–10%

B) 25–30%

C) 70–80%

D) 80–95%

ANSWER: A

127. What is the albedo range for thick clouds?

A) 25–30%

B) 50–60%

C) 70–80%

D) 90–95%

ANSWER: C

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128. What is the albedo range for forests?

A) 5–10%

B) 10–20%

C) 25–30%

D) 35–50%

ANSWER: B

129. What is the albedo range for water bodies, and why does it vary?

A) 1–5%; varies with water depth.

B) 10–60%; varies with the Sun's altitude.

C) 80–95%; varies with ice cover.

D) 20–40%; varies with salinity.

ANSWER: B

130. What wavelength characterizes ultraviolet (UV) radiation compared to the visible spectrum?

A) Longer wavelength than visible light.

B) Shorter wavelength than the visible spectrum.

C) Identical wavelength to infrared radiation.

D) Wavelengths measured in kilometers.

ANSWER: B